

Berger–Hopf Standard Model: Action-domain localization and an auditable route to physical predictions

Norman P. Carberry
Independent Researcher
[ORCID: 0009-0000-6650-3485](https://orcid.org/0009-0000-6650-3485)

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Abstract

The Berger–Hopf Standard Model (BHSM) investigates whether a common stratified variational framework can organize Standard Model-associated structure without using measured masses or mixing angles to choose its upstream dynamics. We formulate the current evidence boundary around action-domain localization, finite-history response, and the transfer from certified mathematical objects to physical observables. The authorized AE3 extension promotes a retained reciprocal-join profile to a response constraint, yielding a regular local material carrier without a new continuous coefficient. Separately, a 512-bit interval calculation composes the frozen central scalar response across 370 intervals and 371 nodes, with maximum response-norm upper bound 8.405509181456809. We describe the complete ambient midpoint chain rule required for the remaining signed transverse calculation and an explicit bound on stored coordinate-solve error. A finite-history local refinement now supplies independently reproduced mean-value enclosures for one trial column; its improvement is distinguished from the still-missing global physical contraction. A separate photon audit corrects a finite-mode symmetry diagnostic while retaining a positive-energy obstruction on the reference Maxwell shell. We also distinguish a stationary Einstein–Cartan domain rejection and a conditional charged-lepton tree sum rule from a physical mass prediction. These results do not establish the joint physical background, particle masses, mixing matrices, scattering observables, or full BHSM completion. The manuscript binds numerical values to immutable source artifacts and identifies the remaining proof operands that must precede a physical prediction.

1 Scientific scope and provenance

BHSM is an evolving mathematical and computational research program. Its historical spectral and coupling screens, conditional structural results, numerical center calculations, and outward certificates have different logical authority. A numerical match cannot close a missing action, domain, normalization, or observable map. The present draft therefore organizes results by their actual hypotheses and reproducible evidence, rather than treating an earlier release label as physical completion.

The current definition of done requires every claimed quantity to share one declared parent action and input ledger, with the intervening variational, operator, scale, and observable maps

closed [2, 3]. Measured mass and mixing data may enter a later comparison; they may not select a branch, coefficient, normalization, regularization, or eigenvector ordering upstream. Frozen historical outputs remain unchanged and retain their original labels.

This manuscript is the working source for the current flagship synthesis. It reports completed local results and a precisely delimited computational frontier. It is not a submission-ready claim of a complete physical model.

2 Stratified action and physical ownership

The retained architecture distinguishes a provisional S^8 parent, a relative two-cap $S^{5|4}$ construction, and an intrinsic localized four-dimensional effective theory. Compatibility maps between these strata are part of the model specification. Fields, coefficients, and domains cannot be moved between strata without an explicit action-owned map.

At the current finite-history frontier, geometry-local derivative kernels are necessary computational inputs. They are not by themselves the complete gauge, ghost, fermion, scalar, and mixed-field derivatives of one physical background. Similarly, an implemented spectrum, vertex, or collision engine does not supply the action-derived operands that it is designed to consume. These distinctions prevent a software capability from being counted as a physical prediction [2, 3].

The current program combines compatible local pieces as they become available. A composed claim is permitted only when its pieces share the action version, background, domain, scale and renormalization conventions, and provenance chain. Later candidate action representatives are not silently identified with an earlier certified representative.

3 The AE3 local material carrier

3.1 Response constraint

The retained reciprocal-join profile is

$$\sigma_0(\chi) = -\frac{1}{2} + \frac{2\chi}{\pi} - \frac{\sin(4\chi)}{2\pi}, \quad 0 \leq \chi \leq \frac{\pi}{2}. \quad (1)$$

AE3 promotes this previously fixed profile into the existing eta-to-sigma response constraint. On the oriented orbit space Q , define

$$W_J[\eta] = \sin^2 f \cos^2 f, \quad Z_J[\eta] = \int_Q W_J[\eta] d\ell, \quad (2)$$

and impose

$$C_\sigma = n_\eta^A \nabla_A \sigma - \frac{W_J[\eta]}{Z_J[\eta]} = 0, \quad \sigma|_{Q_-} = -\frac{1}{2}. \quad (3)$$

The nonpropagating multiplier λ_σ supplies the response term

$$S_{\text{response}} = \int dt \int_Q d\ell \mu_Q \lambda_\sigma C_\sigma. \quad (4)$$

The corresponding action replaces the old fixed profile wherever it occurs and adds this constraint once. It does not add the retained Hopf/FR energy a second time. This authorized action-domain promotion introduces no new continuous coefficient, physical scale, or particle label [4].

3.2 Regularity and scope

On the retained identity branch,

$$\sigma'_0(\chi) = \frac{4}{\pi} \sin^2(2\chi) > 0 \quad (0 < \chi < \pi/2), \quad \sigma_0(\pi/4) = 0, \quad \sigma'_0(\pi/4) = \frac{4}{\pi}. \quad (5)$$

Thus the material zero is transverse. The local carrier is

$$D_{\text{enc}} = \{\sigma < 0\}, \quad \Sigma_{\text{enc}} = \{\sigma = 0\}, \quad n = \frac{\nabla \sigma}{|\nabla \sigma|}. \quad (6)$$

The retained weight $\Lambda(\sigma) = 1 - 4\sigma^2$ is the unique even polynomial of degree at most two with unit value at the seam and zeros at $\sigma = \pm 1/2$. The associated boundary and flux cancellation statements refer to this smooth internal material interface. The event-time condition and the material surface are distinct objects.

The current algebraic transport retains three charged sectors and three family slots. The resulting nine fibers preserve their supplied BHSM state labels through the carrier construction; the geometry does not independently select a particle species. This does not establish commutation with the still-missing interacting history Hamiltonian or a persistent full-field particle [2]. The older unchanged-AE2 audit, in which none of six candidate carriers qualified, remains a valid historical result within AE2.

4 Complete finite-history second variation

4.1 Why the midpoint requires the full ambient space

Let E be the 99-by-74 augmented trial frame, e its current-Green coordinate axis, and V the stored 74-by-73 transverse basis. The retained midpoint tensor contains $H[U, U]$, where $U = EV$ and $H = D^2F$. Transverse endpoint variations need not remain transverse at a Hermite–Simpson midpoint. They can acquire axis and frame-normal components. Dropping those terms is not justified by a small binary64 residual.

For fixed interval length h_i , the endpoint-pair midpoint map has blocks

$$M_L = \frac{1}{2}E_i P_i + \frac{h_i}{8}DF_i E_i P_i, \quad (7)$$

$$M_R = \frac{1}{2}E_{i+1} P_{i+1} - \frac{h_i}{8}DF_{i+1} E_{i+1} P_{i+1}. \quad (8)$$

The fixed reset has zero left variation at interval zero. Complete the actual retained directions to a square ambient basis $S = [U \ C]$, with 26 complement columns, and solve

$$M = [M_L \ M_R] = UA + CD. \quad (9)$$

For each output component, the full pullback is

$$H[M, M] = A^T Q_{UU} A + A^T Q_{UC} D + D^T Q_{UC}^T A + D^T Q_{CC} D. \quad (10)$$

Both cross terms and the complement-complement block are required. Their signs may reinforce or cancel the retained contribution. An incomplete transverse-only norm is neither an upper nor a lower bound for the full pullback [5].

4.2 Signed assembly before norms

Writing $\mathcal{A}_i = -R_i^{-1}E_{i+1}^T$ for the unchanged reduced solve and test-frame map, the local second response is

$$\begin{aligned} \mathcal{S}_i = & \left(\frac{h_i}{6} \mathcal{A}_i + \frac{h_i^2}{12} \mathcal{A}_i D F_{i+1/2} \right) B_i \\ & + \frac{2h_i}{3} \mathcal{A}_i H_{i+1/2} [M_i, M_i] \\ & + \left(\frac{h_i}{6} \mathcal{A}_i - \frac{h_i^2}{12} \mathcal{A}_i D F_{i+1/2} \right) B_{i+1}. \end{aligned} \quad (11)$$

All tensors are first pulled back to the common endpoint-pair coordinates. The midpoint incidence terms retain their signs. If a local block is T and a causal transport is P , the identity

$$\|PT\|_F^2 = \text{tr}(PTT^T P^T) \quad (12)$$

retains output covariance through the transport. Contributions from adjacent intervals acting on the same global diagonal input block must also be added with their signed cross covariance before taking a norm. The existing block-sup proof domain supplies the remaining triangle inequalities.

4.3 Stored-basis and coordinate-solve verification

Binary64 center assembly is distinct from an outward enclosure. For an exact stored binary64 basis S , a 512-bit Arb inverse enclosure verifies nonsingularity without deleting a small singular direction. If $\hat{X}$ is a computed coordinate matrix and M the exact stored binary64 target,

$$\|S^{-1}M - \hat{X}\|_\infty \leq \|S^{-1}\|_\infty \|M - S\hat{X}\|_\infty. \quad (13)$$

Arb supplies the residual and inverse bounds [8, 7]. Equation (13) controls this stored solve only. Physical direction construction, Hessian rounding, propagation of coordinate error through (10), and the action-neighborhood remainder are separate required operands.

5 Verified numerical evidence

The completed correlated scalar calculation uses 512-bit interval arithmetic and the frozen causal frames and preconditioner. Table 1 and Figure 1 are generated from the source certificate after its NPZ data hash is verified [6]. No values are fitted to experimental particle data.

Table 1: Completed frozen central scalar calculation. These are scoped mathematical bounds, not physical mass or scattering observables.

Quantity	Recorded result
Intervals / nodes	370 / 371
Arithmetic precision	512 bits
Maximum causal response norm upper	8.405509181456809
Node owning the maximum	370
Maximum local second-residual norm upper	0.08778167488957692

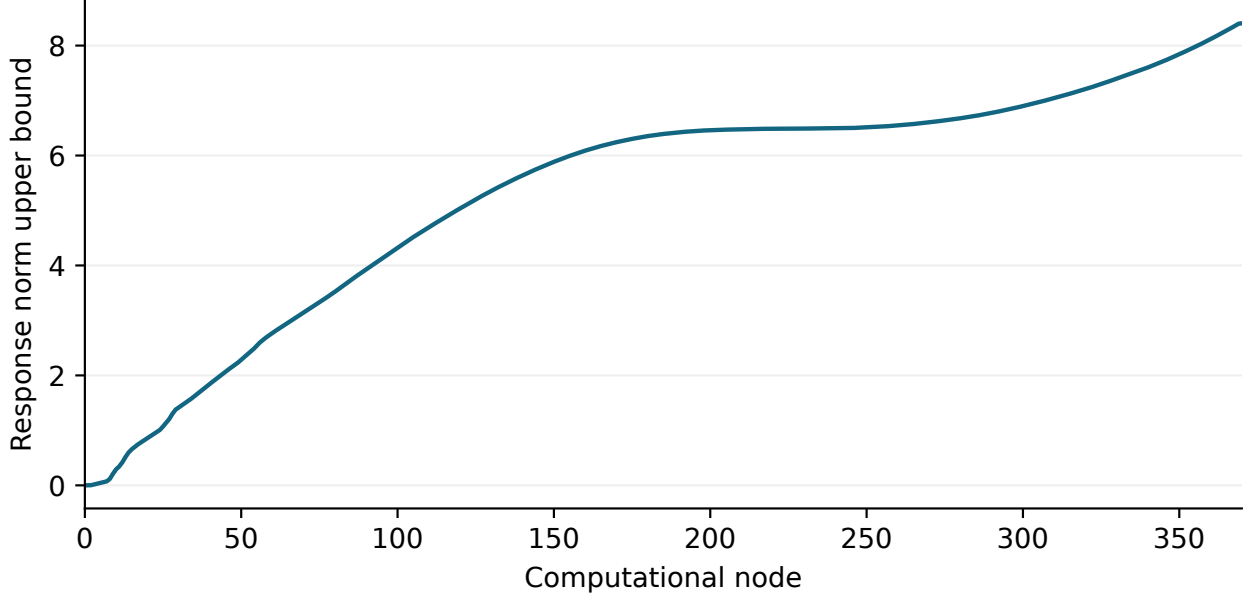


Figure 1: Actual certified numerical response along the frozen path. The horizontal coordinate is a computational node index, not laboratory time. The vertical value is a response-norm upper bound. The central scalar certificate does not enclose the complete transverse neighborhood.

The direct physical curvature campaign covers 371 endpoints and 370 midpoints, followed by the twelve local quadratic source blocks on each of 370 intervals and their signed causal composition. Every point and local source is independently recomputed. Its intended output is a complete fixed-frame center calculation; point coverage alone does not establish a uniform neighborhood bound or the physical quotient [10].

5.1 Paired local mean-value refinement

At interval 13 and frozen trial column 14, paired endpoint and midpoint mean-value calculations retain the original input tubes, all seven bordered solves, scalar contractions, and coupled normalization. The midpoint uses all 249 scaled directions of its actual outer domain. The local integration retains the original exact midpoint center direction, full midpoint first derivative, and center-shift remainder. Table 2 reports the maximum coordinate radii of these enclosures, with inputs generated from the reviewed source artifact [9]. The complete batch reduces the two local

Table 2: Local enclosure refinement at one interval and one trial column. The quantities are interval radii, not measured variations or global operator norms.

Quantity	Before	After
Midpoint center action, left	1.0351495	0.319854856
Midpoint center action, right	1.03810714	0.320674133
Local Newton defect, left	65.6305919	55.9396006
Local Newton defect, right	63.914549	54.9098867

maxima by approximately 15% and 14%. Independent fresh-process runs reproduce their records and arrays byte for byte. The last paired attribution before this refinement identified midpoint

center-action uncertainty as the dominant local term. A subsequent saved-data output-projection probe improved its maximum radius by only about 0.0035%, documenting a limitation of that rearrangement. Neither result establishes full trial coverage or a global physical contraction.

5.2 Self-map headroom and the missing physical certificate

In longitudinal and transverse radii, the nonnegative majorant has the form

$$B_i(r) = Y_i + \sum_j Z_{ij}r_j + C_i r_L^2 + 2M_i r_L r_T + T_i r_T^2, \quad i \in \{L, T\}. \quad (14)$$

The quadratic coefficients include the Taylor factor one half. With compatible uniform bounds on the Newton map and its derivative, sufficient conditions in the radius-weighted maximum norm are

$$r_i - B_i(r) > 0 \quad (i = L, T), \quad \max_i \frac{\sum_j \partial_j B_i(r) r_j}{r_i} < 1. \quad (15)$$

The stored incidence-screen polynomial has trial radii $(9.16018972 \times 10^{-7}, 1.93696126 \times 10^{-9})$. Exact rational evaluation of its stored binary64 coefficients gives positive self-map margins $(2.98613808 \times 10^{-7}, 6.31431658 \times 10^{-10})$ [11]. This screen does not recertify Y , Z , or the central and mixed coefficients and does not include every physical Hessian, frame, or neighborhood error. Its positive headroom is therefore conditional. The complete physical margin remains uncomputed; the local radii in Table 2 cannot be substituted for Z .

The remaining numerical task is to transport compatible uniform errors through the full causal map and demonstrate that their simultaneous costs fit both inequalities. The current widths motivate a uniform enclosure that preserves correlations through the coupled implicit solves and normalized field. Neither greater arithmetic precision nor additional point samples alone supply the missing neighborhood certificate.

6 Photon diagnostic and the retained frozen route

Write the retained dimensionless transverse Dirichlet-to-Neumann kernel as $N(s, q^2)$, with coexact $s = n^2$, $n \geq 2$, and $q^2 = (\omega R_4)^2$. For the normalized regular extension $u(\pi/2) = 1$, put $p = W \sin \rho$, $m = W/\sin \rho$, where $W = 1 - 4\sigma^2$, $\sigma(\rho) = \sigma_0(\rho/2)$, is positive in the interior. The stationary energy is

$$N(s, q^2) = \int_0^{\pi/2} [p(u')^2 + (sm - q^2 p)u^2] d\rho. \quad (16)$$

The regular endpoint makes the integration-by-parts term vanish. At zero frequency, let $T = \int p u^2$, $M = \int m u^2$, and $G = \int p(u')^2$. The variational identities give

$$-\partial_{q^2} N = T, \quad \partial_s N = M, \quad N(s, 0) = G + sM. \quad (17)$$

The earlier complete-mode ratio $sT/N(s, 0)$ is therefore distinct from the spectral-derivative ratio T/M . At $n = 2$, an independent Riccati/sensitivity calculation gives 0.590609601653 and 0.908932991228, respectively [12]. These are binary64 witnesses checked against the retained solver, energy integrals, parameter differences, and cutoff halving; they are not outward enclosures.

The first ratio tests an exact whole-kernel identification with $Z(s - q^2)$, but is not a universal Lorentz-symmetry test for a nonlocal operator. The flat decaying half-space control $N = \sqrt{s - q^2}$ has complete-mode ratio 1/2 and derivative ratio one. Square-root Dirichlet-to-Neumann operators

are standard [14]. Freezing the retained wall coefficients gives that same formal order-one principal symbol on the spacelike branch. This does not construct a global causal quantum propagator or control the null/glancing limit.

A direct obstruction remains on the reference Maxwell shell:

$$N(s, s) = \int_0^{\pi/2} [p(u')^2 + s(m - p)u^2] d\rho > 0. \quad (18)$$

The normalized trace is nonzero and $m - p > 0$ in the interior. At the lowest mode, $N(4, 4) = 0.428281024643$. The isolated frozen kernel has no zero, and its inverse has no pole, on this shell; reflection doubles the positive kernel. This rejects the specified frozen Maxwell route. It does not exclude every completion or establish a global violation of local Lorentz symmetry. Neither diagnostic ratio is a measured speed of light.

7 Stationary domain and conditional flavor evidence

7.1 The Einstein–Cartan candidate

The retained nonzero spin-source mode has $J = \sin^3(2\chi)$, $u_0 = N_0 J^{-1/2} \sin \chi$, and $\Lambda \sim 64\chi^3/(3\pi)$. Its ordinary norm is finite, but

$$\frac{\sin^4 \chi}{J\Lambda} = \frac{3\pi}{512} \chi^{-2} + O(1). \quad (19)$$

This obstruction survives in the uneliminated projected action $\int (AK^2/2 + SK) d\chi$. Compactly supported interior variations require $K_* = -S/A$, with $A = O(\chi^6)$, $S = O(\chi^2)$. The stationary total is $-S^2/(2A) \sim -c\chi^{-2}$, $c > 0$: its quadratic and linear divergences do not cancel. An endpoint condition cannot change that unique interior algebraic solution [13].

For a nonzero pure power-law source mode $u \sim c\chi^\beta$, ordinary normalizability requires $\beta > -2$, while this EC form requires $\beta > -1/4$. The retained $\beta = -1/2$ fails the latter. The result rejects this candidate on the retained mode; it does not repair the action or exclude different backgrounds or vanishing spin sources.

7.2 The lepton relation and its missing dressing

For the admitted heavy, middle, and light charged-lepton slots, $(K, q^2) = (0, 0), (35, 1), (99, 9)$, the retained family formula is

$$m_f = M_{\text{common}} \exp \left[-\frac{K_f + (a^2 - 1)q_f^2}{4\pi} \right]. \quad (20)$$

Eliminating the common scale and squashing gives the exact conditional tree relation

$$R_{\text{tree}} = \log \frac{m_e}{m_\tau} - 9 \log \frac{m_\mu}{m_\tau} - \frac{54}{\pi} = 0. \quad (21)$$

This algebra does not independently select the supplied ledger. For positive dressed masses $M_f = Z_f m_f$, direct substitution gives

$$R_{\text{pole}} = \log Z_e - 9 \log Z_\mu + 8 \log Z_\tau, \quad D = e^{R_{\text{pole}}} = \frac{Z_e Z_\tau^8}{Z_\mu^9}. \quad (22)$$

The stored post-derivation comparison requires $D \simeq 1.06056689915$. It is not an input for choosing a self-energy. A common multiplicative rescaling cancels, but additive or nondiagonal corrections are not excluded. The physical calculation of D and its uncertainty remains open [13].

The family-resolved first-order lepton operator and reset-glued domain already exist. Advanced and retarded Green operators exist on each finite-core globally hyperbolic history; this uses established causal operator theory [15]. It does not select the physical history or Feynman state, evaluate a dressed kernel, or define a global frequency pole on a time-dependent history. The quark response additionally lacks its action-owned up/down vertex normalizations and the left-handed embedding dressing needed for nontrivial physical mixing. These gaps remain distinct from the numerical response campaign.

8 Physical observables and remaining obligations

The project’s physical finish line is stronger than successful local certificates. It requires the complete joint finite-history operator oracle, projected heat-minus-zeta force, KKT root, and constrained physical Hessian, together with compatible downstream maps and continuum control. A positive center screen alone cannot establish these objects.

Every claimed observable must eventually be classified once as a derived point prediction, interval prediction, selection rule, qualitative structure, comparison only, open internal blocker, or outside BHSM 1.0 scope [3]. The required coverage includes the universal electromagnetic vertex and muon magnetic moment; stability and decays; spectral enclosures or exclusions; the known-particle inventory; and representative QED, weak, strong, scalar, flavor, and neutrino scattering benchmarks. This draft does not substitute a list of implemented engines for that completed matrix.

The minimal Standard Model and any effective neutrino extension must remain distinguished. No confinement theorem, completed no-extra-light-state theorem, or first-principles derivation of all Standard Model data is asserted. A failed numerical representation screen also does not prove physical instability or nonexistence of a root.

9 Reproduction, data availability, and manuscript status

The plotted values and the table are generated from the immutable scalar source at revision `d75e77bb`; the downloadable Museum dataset records the full 40-character revision and source SHA-256 digests. The exporter refuses an unvalidated certificate, a changed data hash, a mismatch with the recorded source revision, or an inconsistent array shape or maximum. The plotting generator consumes the same dataset. Repeated artifact materializations are compared byte for byte.

The local table uses revision `68a18555`. The separate local-evidence generator checks that its local-refinement JSON input matches that immutable Git revision. The supplied stored-polynomial snapshot includes all coefficients, radii, exact rational margins, and raw source hashes; its self-map margins are independently replayed with rational arithmetic before writing numerical TeX inputs. This does not verify the missing physical coefficient bounds. No new physical derivatives are evaluated. The local report records raw evidence and reproduction-receipt hashes; exact numerical reproduction additionally requires those retained files and their bound LIVE source bytes, which Git newline normalization must not silently replace.

The complete scientific source and claim ledgers are available in the repository [1]. This draft is maintained separately from frozen historical manuscripts. Its abstract, results, and conclusions

must be updated from completed certificates before external submission; pending physical quantities are not represented by simulated replacement values.

Computational and editorial work used Codex assistance. The scientific provenance is the repository’s retained action, derivations, tests, and certificates. Author declarations, venue-specific formatting, and the final scientific review remain part of submission preparation. No journal submission or acceptance is claimed.

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